



Piscine Rust: Module 00

Summary: ...

Version: 1.00

Contents

I	Preamble	3
II	General Rules	5
III	Exercice 00 : Hello World	6
IV	Exercice 01 : ft_strlen	7

Chapter I

Preamble

Welcome to the first module of the Rust Piscine.

This module serves as an introduction to the `Rust` programming language. Rust is a modern systems programming language designed to provide both high performance and strong memory safety guarantees. It focuses on reliability, efficient resource management, and preventing common classes of programming errors at compile time.

The goal of this module is to help you become familiar with the fundamental concepts of the language.

Throughout this module, you will discover:

- the basic syntax of Rust;
- variables and immutability;
- functions;
- control flow structures;
- the type system;
- the first notions of ownership and borrowing.

Unlike many other programming languages, Rust introduces concepts that may initially seem unfamiliar. This piscine is designed to help you understand them progressively through practice. The exercises encourage experimentation and learning through implementation.

You are encouraged to carefully read the instructions, test your programs frequently, and consult the official documentation when necessary.

Learning Rust requires precision and attention to detail, but it provides powerful tools to write safe, efficient, and maintainable software.

Good luck and enjoy the Rust Piscine.



Take care of your ownership...

Chapter II

General Rules

- Your program should not crash in any circumstances and should not quit unexpectedly. If it happens, your project will be considered non-functional and your grade will be 0.
- Your code must be written in Rust 1.93+
- You must compile your programs with the following command to treat all warnings as errors, ensuring cleaner and safer code: `rustc -D warnings <file.rs>`.
- Your programs must respect clippy linter standards
- Functions names must match exactly what is requested
- Cargo is forbidden only for this module. Only rustc compiler is authorized.
- By Odin, by Thor! Use your brain!!!

Chapter III

Exercice 00 : Hello World

	Exercise 00
Exercise 00 : hello world	
Turn-in directory : <i>ex00/</i>	
Files to turn in : <code>ft_hello.rs</code>	
Allowed functions : <code>print()</code> , <code>println()</code>	

Chapter IV

Exercice 01 : ft_strlen

	Exercise 01
Exercice 01 : ft_strlen	
Turn-in directory : <i>ex01/</i>	
Files to turn in : <code>ft_strlen.rs</code>	
Allowed functions : <code>print()</code> , <code>println()</code> , <code>String</code> method	